

AE-459: Degassing of H2(17)O

Date: 2025-03-11

Tags: Degassed AE Freeze-Pump-Thaw H2(17)O

Category: Two-photon

Status: Done

Created by: Alexander Eith

Reagents

Name	CAS Number / Experiment Number	Inventory number	Amount [mmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Density (g/ml)	Volume [ml]
17O labeled water (10 % 17O)	13768-40-6		/	/	/	/	/	/	apporx. 0.75

Procedure/observations

All steps (unless stated otherwise) were carried out under argon using either standard [Protocol - Schlenk Technique](#) or in an nitrogen filled glovebox (MBRAUN).

Date	Time	Step	Observations
11.03		A 10 mL Schlenk flask was prepared: heat-dried and set under Argon.	
		To the flask H2(17)O was added	
		The water was degassed by freeze-pump-thaw:	
		Approx. 300 mL of acetone were placed in a beaker and liquid nitrogen was added to cool it down, this acetone was used further on and liquid nitrogen was added as needed	
	15:00	The flask was put in a 100 mL beaker filled with acetone (cooled to approx. -5 °C using liquid nitrogen). Thereby the water was stirred at 1000 rpm	
	15:05	After the water solidified the flask was set under vacuum.	
	15:07	The valve to the schlenk line was closed and the acetone bath was removed. The stirring was set to 300 rpm	
	- 15:16	The water was allowed to melt with no bath around it	
	15:17	The flask was put in a 100 mL beaker filled with acetone (cooled to approx. -10 °C using liquid nitrogen). Thereby the water was stirred at 1000 rpm	
	15:27	After the water solidified the flask was set under vacuum.	
	15:28	The valve to the schlenk line was closed and the acetone bath was removed, The stirring was set to 300 rpm	

	- 15:36	The water was allowed to melt with no bath around it	
	15:36	The flask was put in a 100 mL beaker filled with acetone (cooled to approx. -20 °C using liquid nitrogen). Thereby the water was stirred at 1000 rpm	
	15:40	After the water solidified the flask was set under vacuum.	
	15:41	The valve to the schlenk line was closed and the acetone bath was removed. The stirring was set to 300 rpm	
	- 15:50	The water was allowed to melt with no bath around it	
		The flask was set under argon.	
		The flask was stored at rt under argon	

Linked experiments

Prep work - [JU-KRA-173: Activation of molecular sieve](#)

Prep work - [JU-174: Degassing and drying of ACN-d3](#)

Prep work - [AE-409: Degassing and drying of MeCN-d3](#)

Linked resource

Protocol - [Freeze Pump Thaw](#)

Attached files

20250311_145908.jpg

sha256: 5e5e6e8223be4130bc81d1debcda3fcd904ee90bc1309108968878a007b587cb



20250311_145913.jpg

sha256: bf4742061c8da0c4c7e20660b5e9ac05518faf99461e17281ae182ba83c0688a





Unique eLabID: 20250313-246f00aa6df261286fed016b51cd5a1dead15b0e
Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1882>